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Sec: SC-60

Group-17

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PART-A
SINGLE ANSWER TYPE

- Boiling points of hydrogen halides in increasing order
A) HI, HBr, HCl, HF B) HCl, HBr, HI, HF
C) HCl, HF, HBr, HI D) HI, HCl, HBr, HF
- Which one of the following is the strongest oxidising agent?
A) HClO B) HClO₂ C) HClO₃ D) HClO₄
- The solubility of iodine in H₂O may be increased by the addition of
A) Na₂S₂O₃ B) CHCl₃ C) KI D) CS₂
- When chlorine is passed through concentrated hot solution of KOH, the compound formed is
A) KClO B) KClO₂ C) KClO₃ D) KClO₄
- Which one of the following acts as an antichlor?
A) MnO₂ B) Na₂S₂O₃ C) K₂Cr₂O₇ D) Na₂SO₄
- Available chlorine is liberated from bleaching powder when
A) It is heated B) It is reacted with excess of acid solution
C) It is reacted with alkali solution D) It is reacted with water
- Which of the following is the anhydride of perchloric acid?
A) Cl₂O B) ClO₂ C) Cl₂O₆ D) Cl₂O₇
- Which of the following is most stable to heat?
A) HCl B) HOCl C) HBr D) HI
- Chlorine acts as a bleaching agent only in presence of
A) Dry air B) Moisture C) Sunlight D) Pure oxygen
- HI cannot be prepared by the action of conc. H₂SO₄ on KI because
A) HI is stronger than H₂SO₄ B) HI is more volatile than H₂SO₄
C) H₂SO₄ is an oxidising agent D) H₂SO₄ forms complex.
- Oxidising action increases in the following order
A) Cl < Br < I < F B) Cl < I < Br < F
C) I < F < Cl < Br D) I < Br < Cl < F
- Iodine oxidizes the S₂O₃²⁻ ion to
A) SO₃²⁻ B) SO₄²⁻ C) S₄O₆²⁻ D) S²⁻
- A 500 g toothpaste sample has 0.2 g fluoride concentration. What is the concentration of F⁻ ions in terms of ppm level?
A) 250 B) 200 C) 400 D) 1000
- The halogen which is most easily reduced is
A) F₂ B) Cl₂ C) Br₂ D) I₂
- In the manufacture of bromine from sea water, the mother liquor containing bromides is treated with

- A) Carbon dioxide B) Chlorine C) Iodine D) Sulphur dioxide

MULTIPLE ANSWER TYPE

16. Correct statements among the following are
 - A) Iron reacts with hydrochloric acid to form ferrous chloride and not ferric chloride.
 - B) Iodine mono chloride is more reactive than iodine
 - C) Iodine reacts with Chlorine in the presence of water to form iodic acid and hydrogen chloride
 - D) Cyanogen reacts with a base to form Cyanate only.
17. Correct statements are
 - A) I_3^- is linear molecule
 - B) Lone pairs occupy equatorial positions in I_3^-
 - C) I_3^- is not a symmetrical molecule
 - D) It is formed by the interaction between I_2 and I^-
18. HI can be prepared by the following methods
 - A) $PI_3 + H_2O$
 - B) $KI + H_2SO_4$
 - C) $H_2 + I_2 \xrightarrow{Pt}$
 - D) $I_2 + H_2S$
19. Which among the following are diamagnetic?
 - A) Cl_2O
 - B) ClO_2
 - C) Cl_2O_7
 - D) Cl_2O_6
20. Which of the following is/are the characteristics of interhalogen compounds?
 - A) They are more reactive than halogens
 - B) They are quite unstable but none of them is explosive
 - C) They are covalent in nature
 - D) They have low boiling points and are highly volatile
21. Which of the following statements is/are wrong?
 - A) Cl_2 cannot be dried over H_2SO_4 .
 - B) Available chlorine is obtained from caustic soda by treating with HCl.
 - C) Conc. HCl + conc. HNO_3 is Marshall's acid.
 - D) All neutral inter halogen molecules are diamagnetic in nature.
22. Which of the following pair(s) will give chlorine gas most quickly, upon reaction?
 - A) HCl and $KMnO_4$
 - B) NaCl and H_3PO_4
 - C) HCl and MnO_2
 - D) $CaCl_2$ and Br_2
23. Iodine is liberated from sodium iodate by reacting with
 - A) dilute H_2SO_4
 - B) $KMnO_4$
 - C) $NaHSO_3$
 - D) concentrated H_2SO_4 & NaI
24. Which statement is correct about halogens?
 - A) They are all diatomic and form univalent ions
 - B) Halogens have the smallest atomic radii in their respective periods
 - C) They are all diatomic and form diatomic ions
 - D) They are all reducing agents
25. Which of the following pairs is/are correctly matched?
 - A) A halogen which is liquid at room temperature – Bromine
 - B) The most electronegative element–Fluorine
 - C) The most reactive halogen – Fluorine
 - D) The strongest oxidizing agent – Iodine
26. Which of the following will not displace the halogen from the solution of the halide?
 - A) Br_2 added to NaI
 - B) Br_2 added to NaCl
 - C) Cl_2 added to KCl
 - D) Cl_2 added to NaF
27. Identify the incorrect statements among the following.
 - A) Cl_2 reacts with excess of NH_3 to give N_2 and HCl.
 - B) Br_2 reacts with hot and Conc. NaOH solution to give NaBr, $NaBrO_4$ and H_2O .

28. Which one of the following is/are pseudohalide/s?
A) CNO^- B) RCOO^- C) OCN^- D) NNN^-
29. Which of the following is/are oxidised by MnO_2 ?
A) F^- B) Cl^- C) Br^- D) I^-
30. Which is / are true statement(s)?
A) Basic nature of X^- is in order $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$
B) HI is strongest acid of HF, HCl, HBr and HI
C) The ionic character of $\text{M}-\text{X}$ bond decreases in the order $\text{M}-\text{F} > \text{M}-\text{Cl} > \text{M}-\text{Br} > \text{M}-\text{I}$
D) Among ions of F, Cl, Br and I, F has the highest enthalpy of hydration.

PART-B

SINGLE ANSWER TYPE

- Iodine oxide used in estimation of carbon monoxide is
A) I_2O_4 B) I_2O_7 C) I_2O_5 D) I_2O_6
- Which acid can combine with its own salt again?
A) HF B) HBr C) HCl D) HI
- Cupric salts react with potassium iodide to produce
A) Copper and Iodine B) Cuprous salt and Iodate ion
C) Cuprous salt and Iodine D) Copper and Iodate ion
- Heat of vaporization is highest in case of
A) I_2 B) Cl_2 C) Br_2 D) F_2
- Impurity present in Caliche contains one of the following halogen
A) Fluorine B) Bromine C) Iodine D) Chlorine
- Euchlorine is a mixture of
A) Chlorine and Chlorine dioxide B) Chlorine dioxide and Potassium chloride
C) Chlorine gas and Potassium chlorite D) Chlorine gas and Chlorine trioxide.
- When dry chlorine is passed through silver chlorate heated to 90°C then the oxide of chlorine obtained is
A) ClO_3 B) Cl_2O_7 C) ClO_2 D) Cl_2O
- With limited supply of NH_3 , F_2 gives
A) NF_3 B) HN_3 C) N_2 D) N_2H_4
- Among the fluorides given below which will further react with F_2 ?
A) NaF B) CaF_2 C) SF_6 D) IF_5
- When fluoride is heated with conc. H_2SO_4 and MnO_2 the gas evolved is
A) HF B) F_2 C) SF_4 D) None
- When Cl_2 is passed through hot and conc. solution of KOH, the compound formed is
A) KClO B) KClO_3 C) KClO_2 D) KClO_4
- Iodine displaces chlorine from which one of the compounds?
A) KCl B) CaCl_2 C) CCl_4 D) KClO_3
- Which of the following polyhalides is not known?
A) F_3^- B) Cl_3^- C) Br_3^- D) I_3^-

14. Which of the following is a pseudohalogen?
 A) IF_7 B) $(\text{CN})_2$ C) ICl_2 D) I_3^-
15. The structure of ClF_3 is
 A) Pentagonal bipyramid B) Trigonal bipyramid
 C) Pentagonal bipyramid D) T-shape
16. Bromine can be liberated from potassium bromide solution by the action of
 A) Iodine solution B) Chlorine water
 C) Sodium chloride D) Potassium iodide
17. Bleaching powder is obtained by the action of chlorine gas and
 A) Dilute solution of $\text{Ca}(\text{OH})_2$ B) Concentrated solution of $\text{Ca}(\text{OH})_2$
 C) Dry CaO D) Dry slaked lime
18. Conc. HNO_3 reacts with iodine to give
 A) HI B) HOI C) HOIO_2 D) HOIO_3
19. Which one of the following sequences represents the correct increasing order of bond angles in the given molecules?
 A) $\text{H}_2\text{O} < \text{OF}_2 < \text{OCl}_2 < \text{ClO}_2$ B) $\text{OCl}_2 < \text{ClO}_2 < \text{H}_2\text{O} < \text{OF}_2$
 C) $\text{OF}_2 < \text{H}_2\text{O} < \text{OCl}_2 < \text{ClO}_2$ D) $\text{ClO}_2 < \text{OF}_2 < \text{OCl}_2 < \text{H}_2\text{O}$
20. The correct sequence of arrangement of the following compounds in order of decreasing oxidation numbers of iodine is
 A) $\text{HIO}_4, \text{HI}, \text{I}_2, \text{ICl}_5$ B) $\text{HIO}_4, \text{ICl}_5, \text{HI}, \text{I}_2$
 C) $\text{ICl}_5, \text{HIO}_4, \text{I}_2, \text{HI}$ D) $\text{HIO}_4, \text{ICl}_5, \text{I}_2, \text{HI}$
21. Among the following, which is the strongest reducing agent?
 A) F_2 B) Cl_2 C) Br_2 D) I_2
22. The following acids have been arranged in the order of decreasing acid strength. Identify the correct order. ClOH (I) BrOH (II) IOH (III)
 A) $\text{I} > \text{II} > \text{III}$ B) $\text{II} > \text{I} > \text{III}$ C) $\text{III} > \text{II} > \text{I}$ D) $\text{I} > \text{III} > \text{II}$
23. Which of the following can give iodometric titration?
 A) Fe^{2+} B) Cu^{2+} C) Pb^{2+} D) Ag^{2+}
24. In the reaction $\text{I}_2 + \text{I}^- \longrightarrow \text{I}_3^-$ the Lewis base is
 A) I^- B) I_2 C) I_3^- D) None of these
25. The reaction of KMnO_4 and HCl results in
 A) Oxidation of Mn in KMnO_4 and production of Cl_2
 B) Reduction of Mn in KMnO_4 and production of H_2
 C) Oxidation of Mn in KMnO_4 and production of H_2
 D) Reduction of Mn in KMnO_4 and production of Cl_2
26. The reaction $3\text{ClO}^-(\text{aq}) \longrightarrow \text{ClO}_3^-(\text{aq}) + 2\text{Cl}^-(\text{aq})$ is an example of?
 A) Oxidation reaction B) Reduction reaction
 C) Disproportionation reaction D) Decomposition reaction
27. Which one of the following order is correct for the bond energies of halogen molecules?
 A) $\text{I}_2 > \text{Cl}_2 > \text{Br}_2$ B) $\text{Br}_2 > \text{Cl}_2 > \text{I}_2$ C) $\text{I}_2 > \text{Br}_2 > \text{Cl}_2$ D) $\text{Cl}_2 > \text{Br}_2 > \text{I}_2$
28. The manufacture of fluorine is done by
 A) Heating anhydrous HF and MnO_2 .
 B) Electrolysis of aqueous HF .
 C) Electrolysis of anhydrous HF mixed with KHF_2 .
 D) Heating a mixture of KF , MnO_2 and conc. H_2SO_4 .
29. Which halogen oxidises water to liberate oxygen exothermally?

- A) Fluorine B) Chlorine C) Bromine D) Iodine
30. Fluorine is obtained by the interaction of K_2MnF_6 with Lewis acid SbF_5 because of the:
- A) Acidolysis of MnF_4 B) Decomposition of MnF_4
 C) Ionization of MnF_4 D) Decomposition of SbF_6

PART-C

SINGLE ANSWER TYPE

1. $Cl_2(g) + Ba(OH)_2 \longrightarrow X + BaCl_2 + H_2O$
 $X + H_2SO_4 \longrightarrow Y + BaSO_4$
 Potassium salt of Y liberates a gas "Z" on heating in presence of MnO_2 , then Z is?
 A) ClO_2 B) Cl_2O C) O_3 D) O_2
2. 2 g of bleaching powder is dissolved in 100 mL of water, 20 mL of this solution were heated with excess of CH_3COOH and KI and then titrated with 0.12 N hypo solution. In this titration 25 mL of hypo was used. The percentage of available chlorine in this sample of bleaching powder.
 A) 26.25 B) 35.55 C) 12.5 D) 48.10
3. Hypo halite of chlorine in basic medium gives
 A) Halate ion and Chlorine gas B) Halate ion and Per halate ion
 C) Halate ion and Chloride ion D) Halite ion and Chloride ion
4. Which reaction yields the greatest quantity of chlorine from a given quantity of hydrochloric acid?
 A) $MnO_2 + HCl \xrightarrow{\Delta}$ B) $PbO_2 + HCl \xrightarrow{\Delta}$
 C) $KMnO_4 + Conc.HCl \longrightarrow$ D) $CaOCl_2 + HCl \longrightarrow$
5. Oxide of chlorine which is paramagnetic in liquid state and diamagnetic in solid state
 A) Cl_2O_6 B) ClO_2 C) Cl_2O D) Cl_2O_7
6. A dark violet solid X reacts with NH_3 to form a mild explosive which decomposes to give a violet coloured gas. X also reacts with H_2 to give an acid Y. Y can also be prepared by heating its salt with H_3PO_4 . X and Y are
 A) Cl_2 , HCl B) SO_2 , H_2SO_4 C) Br_2 , HBr D) I_2 , HI
7. Chlorine is liberated when we heat
 A) $KMnO_4 + NaCl$ B) $K_2Cr_2O_7 + MnO_2$
 C) $Pb(NO_3)_2 + MnO_2$ D) $K_2Cr_2O_7 + HCl$
8. Concentrated H_2SO_4 cannot be used to prepare HBr from NaBr, because it
 A) Reduces HBr B) Oxidises HBr
 C) Disproportionates HBr D) Reacts slowly with NaBr

MULTIPLE ANSWER TYPE

9. Select INCORRECT order(s) from the following
 A) Stability: $ClO^- < ClO_2^- < ClO_3^- < ClO_4^-$
 B) Bond length: $ClO^- < ClO_2^- < ClO_3^- < ClO_4^-$
 C) Acid strength: $HClO < HClO_2 < HClO_3 < HClO_4$
 D) Oxidizing power: $HClO < HClO_2 < HClO_3 < HClO_4$
10. Select correct statement(s)

- A) ClO_2 and Cl_2O are used as bleaching agents for paper pulp and textiles.
 B) OCl^- disproportionates in alkaline medium.
 C) BrO_3^- liberates Br_2 with iodine in acidic medium.
 D) HClO_2 liberates iodine from KI .
11. What products are expected from disproportionation reaction of hypochlorous acid?
 A) HClO_3 B) HClO_2 C) HCl D) HClO_4
12. Select the correct order(s).
 A) $\text{HOCl} > \text{HOBr} > \text{HOI}$ – Acid strength.
 B) $\text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2 < \text{HClO}$ – oxidising power.
 C) $\text{ClO}_4^- < \text{BrO}_4^- < \text{IO}_4^-$ – oxidising power.
 D) $\text{IO}^- > \text{BrO}^- > \text{ClO}^-$ – ease of disproportionation.
13. Which amongst the following reactions can be used for the respective preparation?
 A) $2\text{KBr} + \text{H}_2\text{SO}_4 (\text{Conc.}) \longrightarrow \text{K}_2\text{SO}_4 + 2\text{HBr}$
 B) $\text{NaCl} + \text{H}_2\text{SO}_4 (\text{Conc.}) \longrightarrow \text{NaHSO}_4 + \text{HCl}$
 C) $\text{NaHSO}_4 + \text{NaCl} \longrightarrow \text{Na}_2\text{SO}_4 + \text{HCl}$
 D) $\text{CaF}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{CaSO}_4 + 2\text{HF}$
14. Select correct statements
 A) Cl_2O and ClO_2 are used as bleaching agents and germicides.
 B) I_2O_5 is used in the quantitative estimation of CO .
 C) Cl_2O explodes in presence of ammonia forming NH_4Cl and liberating N_2 gas.
 D) ClO_2 is paramagnetic in nature
15. Which of the following product(s) is/are obtained when Cl_2O_6 reacts with KOH ?
 A) KCl B) KClO_2 C) KClO_3 D) KClO_4
16. Which of the following is/are incorrect statements?
 A) F_2 has higher dissociation energy than Cl_2
 B) F has higher electron affinity than Cl
 C) HF is stronger acid than HCl
 D) Boiling point increases down the group in halogens

PASSAGE-1

A red liquid (A) when treated with sodium carbonate gives a mixture of two salts (B) and (C) in the solution. The mixture then on acidification with sulphuric acid and distillation produces the red liquid (A) again.

17. Select the correct statement for the liquid (A).
 A) It acts as an oxidising agent
 B) It is sparingly soluble in water
 C) It converts the yellow-dye stuff fluorescence (I) into red colour compound
 D) All of these
18. Which of the following statement is false for salt (B)?
 A) Its solution in water gives pale yellow precipitate with silver nitrate solution
 B) Its solution in water gives white precipitate with lead nitrate solution
 C) Its acidified solution (with conc. H_2SO_4) liberates a coloured gas which produces orange red spots on starch paper
 D) Action of I_2 on aqueous solution of salt (B) produces gas A
19. Which of the following statement is correct?
 A) Liquid (A) undergoes disproportionation reaction in aqueous solution of sodium carbonate
 B) The anion of compound (C) has sp^3 hybridisation and is trigonal pyramidal in shape

- C) Both (A) and (B)
D) None of these

PASSAGE-2

White crystalline solid (A) reacts with H_2 to form a highly associated liquid (B) and a monoatomic, colorless gas (C). The liquid (B) is used for etching glass. Compound (A) undergoes hydrolysis slowly to form (C), (B) and a diatomic gas (D) whose IE is almost similar to that of (C). (B) forms an addition compound with KF to form (E) which is electrolysed in the molten state to form a most reactive gas (F) which combines with (C) in 2 : 1 ratio to produce (A).

20. According to Molecular Orbital Theory, which of the following is correct about the molecule D?
A) Its bond order is 2.0 B) It has two unpaired electrons in π -bonding M.O.
C) Both the above are correct D) None of these is correct
21. Which of the following is correct for the white crystalline solid (A)?
A) It oxidises F^- to F_2
B) It on hydrolysis with alkali under goes disproportionation.
C) It is obtained by the reaction of (C) with O_2F_2 at $118^\circ C$.
D) It acts as strong reducing agent.
22. The compound 'A' reacts with sulphur to form a compound in which hybridisation state of sulphur atom is
A) sp^3d B) sp^3d^2 C) sp^3 D) sp^3d^3

MATRIX MATCH TYPE

23. Match the reactions listed in column-I with the product(s) listed in column-II.

COLUMN - I	
A)	$Cl_2O_6 + H_2O$
B)	$NaClO_4(s) + HCl(conc.)$
C)	$KClO_3 + (COOH)_2$
D)	$HgO + Cl_2$

COLUMN - II	
P)	ClO_2
Q)	$HClO_3$
R)	Cl_2O
S)	$HClO_4$

24. Match the reactions listed in column-I with the product(s) listed in column-II.

COLUMN - I	
A)	F_2
B)	Cl_2
C)	Br_2
D)	I_2

COLUMN - II	
P)	Decomposition of water
Q)	Formation of poly halides
R)	Extraction of gold & platinum
S)	Germicide

25. Match the reactions listed in column-I with the product(s) listed in column-II.

COLUMN - I	
A)	Maximum solubility in water
B)	Corrosive liquid
C)	Maximum intermolecular distance

COLUMN - II	
P)	F_2
Q)	Cl_2
R)	Br_2

D) Maximum Enthalpy of dissociation | S) I₂

KEY

PART-A

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|---------|----------|----------|---------|---------|---------|---------|
| 1. B | 2. A | 3. C | 4. C | 5. B | 6. B | 7. D |
| 8. A | 9. B | 10. C | 11. D | 12. C | 13. C | 14. A |
| 15. B | 16. ABC | 17. ABCD | 18. ACD | 19. ACD | 20. ABC | 21. ABC |
| 22. AC | 23. CD | 24. AB | 25. ABC | 26. BCD | 27. AB | 28. ACD |
| 29. BCD | 30. ABCD | | | | | |

PART-B

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|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. A | 3. C | 4. A | 5. C | 6. A | 7. C |
| 8. C | 9. D | 10. A | 11. B | 12. D | 13. A | 14. B |
| 15. D | 16. B | 17. D | 18. C | 19. C | 20. D | 21. D |
| 22. A | 23. B | 24. A | 25. D | 26. C | 27. D | 28. C |
| 29. A | 30. B | | | | | |

PART-C

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|------------------------|-------------------------|----------|--------|----------------------------|---------|----------|
| 1. D | 2. A | 3. C | 4. D | 5. A | 6. D | 7. D |
| 8. B | 9. BD | 10. ABCD | 11. BC | 12. ABD | 13. BCD | 14. ABCD |
| 15. CD | 16. ABC | 17. D | 18. D | 19. C | 20. A | 21. C |
| 22. B | 23. A→QS, B→S, C→P, D→R | | | 24. A→P, B→PQRS, C→PQ, D→Q | | |
| 25. A→P, B→R, C→S, D→Q | | | | | | |